

Ankit

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Summary

M.Sc. candidate in Computational Mathematics and Data Science at NIT Rourkela with working knowledge in Python, C++, statistics, machine learning basics and deep learning basics. Academic background covers linear algebra, probability, numerical analysis, and optimization.

Education

National Institute of Technology, Rourkela	Aug 2025 – Present
M.Sc. in Computational Mathematics and Data Science (Certificate)	CGPA: 7.95
Sh. Rameshwar Lal Sharma Teacher Training College, Pilani	2021 – 2025
B.Sc. – B.Ed (Certificate)	CGPA: 7.23

Research Experience

Numerical Analysis Laboratory	Aug 2025 – Nov 2025
Python, NumPy, SciPy	

- Implemented algorithms for nonlinear equations, linear systems, and ODE models
- Performed convergence and error analysis for numerical stability
- Benchmarked efficiency of iterative computational methods

Statistics Laboratory	Jan 2026 – Present
C++	

- Built statistical computation programs from scratch (no STL)
- Implemented simulation models for probability distributions
- Computed descriptive and inferential statistics efficiently.

Projects

Heatwave Prediction in Rajasthan	
Python, FastAPI, Scikit-learn, Pandas, NumPy, Docker	

- Engineered a production-ready ML system to predict heatwave conditions and assess outdoor work safety, achieving 92.5% classification accuracy.
- Developed 5 RESTful API endpoints using FastAPI with Pydantic validation for real-time predictions, weather fetching, and risk assessment.
- Built feature engineering pipeline incorporating temperature-humidity interactions (heat index) and auto-fetched OpenWeather API data.
- Deployed RandomForestClassifier generating risk-stratified outputs (Low/Medium/High) with binary heatwave predictions and work advisories.
- Containerized with Docker for reproducible deployment; integrated frontend UI for end-to-end prediction workflow.

Sleep Breathing Analysis	
Python, TensorFlow, Keras, NumPy, SciPy, Matplotlib	

- Built an end-to-end pipeline to classify sleep breathing events (Hypopnea, Obstructive Apnea) from PSG signals across 5 participants.
- Applied 4th-order Butterworth bandpass filter (0.17–0.4 Hz), segmented into 30-second windows with 50% overlap, and trained a 1D CNN evaluated via Leave-One-Participant-Out Cross-Validation with classification reports and confusion matrices.
- Generated per-participant PDF diagnostic reports with annotated PSG signal overlays.

Technical Skills

Programming: Python | **APIs:** FastAPI
Data Analysis: Pandas, NumPy, Matplotlib-seaborn visualization, EDA, Statistical Modeling
Machine Learning: Regression, Decision Trees, ensemble, Model Evaluation
Deep Learning: TensorFlow, Keras, 1D CNN
Database: MySQL **Tools:** Git, GitHub

Certification

Python Essentials – Cisco (2025) (Certificate)